

VASIREDDY ABHISHEK SAI

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PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building LLM pipelines, RAG systems, multi-agent workflows, and computer vision models. Final-year B.Tech (AI & ML) student at Manipal Institute of Technology with 2 internships and 6 shipped projects across NLP, generative AI, medical imaging, and production MLOps. Comfortable working across the full ML lifecycle — from data pipelines and model training to deployment and monitoring. Open to product and service environments, with experience in client-facing analytics and reporting.

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace Transformers, OpenCV, Sentence Transformers, ONNX

Generative AI & LLMs: LangChain, LangGraph, LlamaIndex, RAG, LLM Fine-tuning (LoRA, QLoRA, PEFT), Prompt Engineering, Multi-Agent Systems, OpenAI API, Anthropic API, vLLM, Ollama

MLOps & Infra: MLflow, Docker, FastAPI, Streamlit, Prometheus, Grafana, Evidently AI, GitHub Actions, RAGAS, DVC

Vector & Search: FAISS, ChromaDB, Pinecone, BM25, Hybrid Search, Cohere Re-ranking

Cloud & Data: AWS (SageMaker, Bedrock), Azure OpenAI, GCP Vertex AI, PostgreSQL, MongoDB, Apache Airflow, Kafka, PySpark, n8n

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern — Incresol Software Services Pvt. Ltd., Hyderabad

January 2026 – Present

- Built LangChain-based automation workflows with tool-use and memory management that cut report turnaround time by **60%** — the main bottleneck was unstructured agent outputs, which I fixed by adding structured output parsing and a retry layer.
- Designed and deployed **12–15 Airflow and n8n data pipelines** that replaced manual CSV exports across 3 teams, saving **15–20 person-hours per week**.
- Built a PostgreSQL-to-MongoDB sync pipeline that processes **100K+ records monthly** — handled schema mismatches and partial failures using idempotent writes and a dead-letter queue.
- Optimized slow procurement queries using index tuning and materialized views; queries that took 8–10 seconds now run in under 2 seconds. Also built Streamlit dashboards for KPI visibility.
- Mentored 2 junior engineers on pipeline design; documented architecture decisions and presented system demos to stakeholders across 4 sprints.

AI/ML Engineer Intern — BRC Web, Hyderabad

May 2025 – August 2025

- Integrated OpenAI API with LangChain to automate content generation for client reports — reduced production time by **60%** mostly by replacing manual templating with structured prompt pipelines.
- Trained Random Forest, XGBoost, and Logistic Regression models for customer churn prediction; best model hit **88%+ accuracy** in production. Deployed predictions via a FastAPI REST endpoint.
- Wrote Python and SQL ETL pipelines to clean and standardize raw client data — cut data preparation time by **40%** and fixed several recurring feature engineering bugs.
- Built spaCy and HuggingFace sentiment analysis pipeline on **50K+ customer reviews**; surfaced insights in Power BI and Metabase dashboards shared with the product team weekly.

KEY PROJECTS

Production RAG Pipeline | github.com/abhishekvasireddy/production-rag-pipeline

- Built a RAG system that serves **10K+ queries/day** using hybrid BM25 + dense retrieval and Cohere re-ranking — improved top-5 retrieval accuracy by **34%** vs dense-only.
- Had a latency problem early on (2s+ responses) — switched from sequential to async retrieval with FastAPI and FAISS, bringing p95 latency down to **under 800ms**.
- Wrote a RAGAS evaluation harness that measures faithfulness, answer relevance, and context precision on **500+ test queries** and runs automatically in CI on every merge.

ML Monitoring & Drift Detection | github.com/abhishekvasireddy/ml-monitoring-drift-detection

- Built a monitoring stack using Prometheus, Grafana, MLflow, and Evidently AI that tracks **10+ production models** for data drift and performance degradation in real time.
- Implemented PSI and KL-divergence checks on feature distributions; when drift crosses a threshold, the system auto-triggers a retraining workflow and sends an alert.
- Packaged everything in Docker with GitHub Actions CI/CD — mean time to detect model decay went from **days to under 15 minutes**.

AI Agents for Medical Diagnostics | github.com/abhishekvasireddy/AI-Agents-for-Medical-Diagnostics

- Built a multi-agent system with 3 specialized LLM agents — symptom intake, differential diagnosis, and triage — orchestrated through LangChain with shared memory and tool-use.
- Added clinical guardrails and structured function calling per agent to prevent hallucination; each diagnosis includes a confidence score and reasoning trace.
- Evaluated against a benchmark of **200+ clinician-labeled cases** and achieved **91% top-3 diagnostic agreement**.

Multimodal Document Intelligence | github.com/abhishekvasireddy/multimodal-document-intelligence

- Built a document Q&A system that handles PDFs, scanned images, and mixed-format files — uses OCR for layout extraction and a vision-language model for understanding.
- Answers are grounded in source documents using ChromaDB + Sentence Transformers; each response includes a bounding-box citation so users can verify the source.
- Tested on **5,000 documents** (invoices, contracts, forms) — cut manual review time by **~70%** in internal testing.

AI Cervical Cancer Detection | github.com/abhishekvasireddy/AI-Cervical-Cancer-Detection

- Trained an EfficientNet-B3 classifier on the SIPaKMeD dataset to categorize cervical cells into 5 classes for early cancer detection — achieved **96.4% test accuracy** and 0.95 macro-F1.
- Dataset was class-imbalanced; fixed using stratified k-fold CV and augmentation — without this, accuracy dropped to ~88%.
- Exported the model to ONNX and wrapped it in a FastAPI service; inference runs at **under 100ms per image** on CPU.

LLM Fine-Tuning — Procurement Domain | github.com/abhishekvasireddy/llm-finetune-procurement

- Fine-tuned a 7B LLM on procurement data (POs, PRs, vendor spend) using QLoRA via PEFT — trained on a single A100, cut trainable parameters by **98%** vs full fine-tune.
- Built the full pipeline: dataset curation, instruction tuning, RAGAS evaluation, and GGUF quantization for CPU inference via Ollama.
- Domain Q&A accuracy improved by **27%** over the base model on a held-out test set of **1,200 procurement prompts**.

EDUCATION

B.Tech — Computer Science Engineering (Artificial Intelligence & Machine Learning) | Manipal Institute of Technology, Manipal | 2021 – 2027

CERTIFICATIONS

- Google AI Essentials & Generative AI — Google / Coursera (March 2026)
- Machine Learning with Python — IBM / Coursera (April 2026)
- Google Data Analytics Professional Certificate — Google / Coursera (November 2025)
- Deep Learning Specialization — DeepLearning.AI, Andrew Ng (February 2025)
- Machine Learning Specialization — DeepLearning.AI, Andrew Ng (August 2024)